Ayrık Zamanlı Fourier Serileri

(Discrete-Time Fourier Series)

$x[n] = x[n+N]$, $\forall n$, $N = \frac{2\pi}{\omega_0}$, $\omega_0 = \frac{2\pi}{N}$

Periyodik bir işaret Fourier serisine açılabilir.

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}$$

$$a_k = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jk\omega_0 n}$$

⇒ Ayrık zamanlı periyodik bir işaretin Fourier seri açılımı sonlu seri olup, yakınsaklık problemi yoktur.

Reminder: Ayrık zamanlı karmaşık üstel işaretler ($e^{j\omega_0 n}$) 2π'nin katları olan frekanslarda aynıdır.

Temel fonksiyon: $e^{jk\omega_0 n} = \psi_k[n]$, $k = 0, \pm 1, \pm 2, \ldots$

$\psi_k[n]$, $\omega_0 = 2\pi/N$ temel frekansının $k\omega_0$ şeklinde katları olan frekanslara sahip olduğundan harmonik olarak ilişkilidir ve bunlarda N farklı işaret vardır.

$\psi_k[n] = \psi_{k+N}[n]$

İspat: $e^{jk\omega_0 n} \stackrel{?}{=} e^{j(k+N)\omega_0 n} = e^{jk\omega_0 n} \cdot e^{jN\omega_0 n} = e^{jk\omega_0 n} \cdot e^{j2\pi n} = e^{jk\omega_0 n}$

($N\omega_0 = 2\pi$, $e^{j2\pi n} = 1$)

$$\Rightarrow \sum_{k=<N>} e^{jk\omega_0 n} = \begin{cases} N, & k = 0, \pm N, \pm 2N, \ldots \\ 0, & \text{diğerinde} \end{cases}$$

Karmaşık üstel işaretler sabit olmadıkça bir periyottaki değerlerinin toplamı sıfırdır.

İspat:

$$\sum_{n=q}^{q+N-1} e^{jk\omega_0 n} = \sum_{n=0}^{N-1} e^{jk\omega_0 (q+n)} = e^{jk\omega_0 q} \sum_{n=0}^{N-1} e^{jk\omega_0 n}$$

$$= e^{jk\omega_0 q} \left[\frac{1 - e^{jk\omega_0 N}}{1 - e^{jk\omega_0}} \right] = 0 \quad , \quad k \neq pN$$

($\omega_0 N = 2\pi$)

$$k = pN \rightarrow \sum_{n=q}^{q+N-1} e^{jpN\omega_0 n} = \sum_{n=q}^{q+N-1} e^{j2\pi pn} = \sum_{n=q}^{q+N-1} 1 = N$$

⇒ $x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}$ ← $e^{-jr\omega_0 n}$ çarpıp, N terimli toplam

$$\sum_{n=<N>} x[n] e^{-jr\omega_0 n} = \sum_{n=<N>} \sum_{k=<N>} a_k e^{jk\omega_0 n} \cdot e^{-jr\omega_0 n} = \sum_{k=<N>} a_k \underbrace{\sum_{n=<N>} e^{j(k-r)\omega_0 n}}_{\begin{cases} N, & k=r \\ 0, & \text{diğerinde} \end{cases}} = a_r \cdot N$$

$$a_r = \frac{1}{N} \sum_{n=<N>} x[n] e^{-jr\omega_0 n}$$

⇒ Ayrık zamanlı Fourier serisi açılımında, FS katsayıları x_k'larda N periyodu ile periyodiktir.

$x_k = x_{k+N}$

İspat: $x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n} = \sum_{k=<N>} x_k \psi_k[n]$

$= x_0\psi_0[n] + x_1\psi_1[n] + \cdots + x_{N-1}\psi_{N-1}[n]$ ⟵ $k = 0, 1, 2, \ldots, N-1$

$= x_1\psi_1[n] + \cdots + x_{N-1}\psi_{N-1}[n] + x_N\psi_N[n]$ ⟵ $k = 1, 2, \ldots, N$

$\psi_0[n] = \psi_N[n]$ olduğundan $x_0 = x_N$ olmalıdır.

Ex: $x[n] = \sin 3\omega_0 n$ işaretinin FS katsayılarını bulunuz ve N=5 için Fourier spektrumlarını çiziniz.

$x[n] = \frac{1}{2j}\left[e^{j3\omega_0 n} - e^{-j3\omega_0 n}\right]$

$x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n}$

$k = \mp 3 \longrightarrow x_3 = \frac{1}{2j} = -j/2$

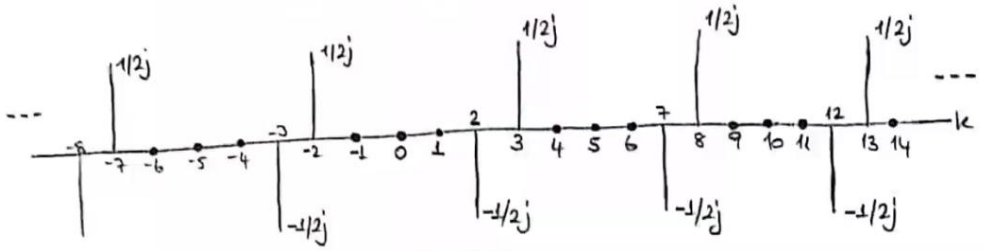
$x_{-3} = -\frac{1}{2j} = j/2$

$x_k = |x_k| e^{j\theta_k} \longrightarrow x_3 = -j/2 \longrightarrow |x_3| = 1/2$ ve $\theta_3 = -\pi/2$

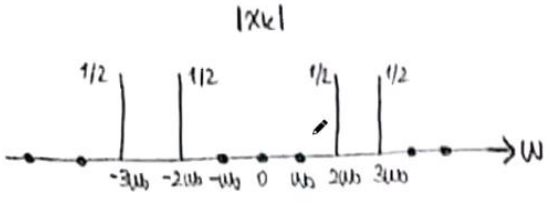
$\hookrightarrow x_{-3} = j/2 \longrightarrow |x_{-3}| = 1/2$ ve $\theta_{-3} = \pi/2$

$x_k = x_{k+rN} \xrightarrow[N=5]{} x_3 = x_8 = x_{13} = \cdots$

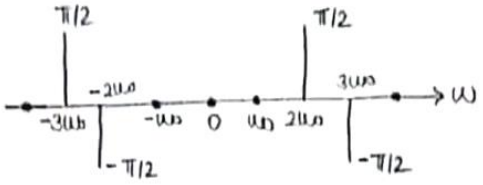
$x_{-3} = x_2 = x_7 = \cdots$



x_k, N=5



Genlik spektrumu



Faz spektrumu

Ex: $x[n] = 1 + \sin\left(\frac{2\pi}{N}n\right) + 3\cos\left(\frac{2\pi}{N}n\right) + \cos\left(\frac{4\pi}{N}n + \frac{\pi}{2}\right)$

$\omega_0 = 2\pi/N$

$$x[n] = 1 + \frac{1}{2j}\left[e^{j\omega_0 n} - e^{-j\omega_0 n}\right] + \frac{3}{2}\left[e^{j\omega_0 n} + e^{-j\omega_0 n}\right] + \frac{1}{2}\left[e^{j(2\omega_0 n + \pi/2)} + e^{-j(2\omega_0 n + \pi/2)}\right]$$

$x_0 = 1$

$x_1 = \frac{1}{2j} + \frac{3}{2} = \frac{3}{2} - \frac{1}{2}j$

$x_{-1} = -\frac{1}{2j} + \frac{3}{2} = \frac{3}{2} + \frac{1}{2}j$

} $x_1^* = x_{-1}$

$\longrightarrow |x_1| = \sqrt{(3/2)^2 + (1/2)^2} = \sqrt{10}/2$

$\theta_1 = \tan^{-1}\left(\frac{-1/2}{3/2}\right) = -\tan^{-1}(1/3)$

$\longrightarrow |x_{-1}| = \sqrt{10}/2$

$\theta_{-1} = \tan^{-1}(1/3)$

$x_2 = \frac{1}{2}e^{j\pi/2} = \frac{1}{2}j$

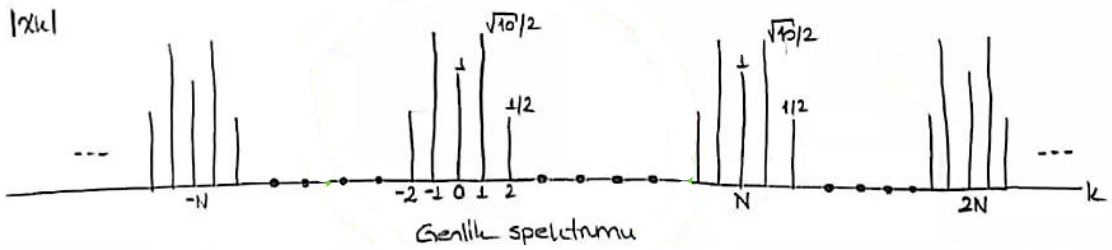
$x_{-2} = \frac{1}{2}e^{-j\pi/2} = -\frac{1}{2}j$

} $x_2^* = x_{-2}$

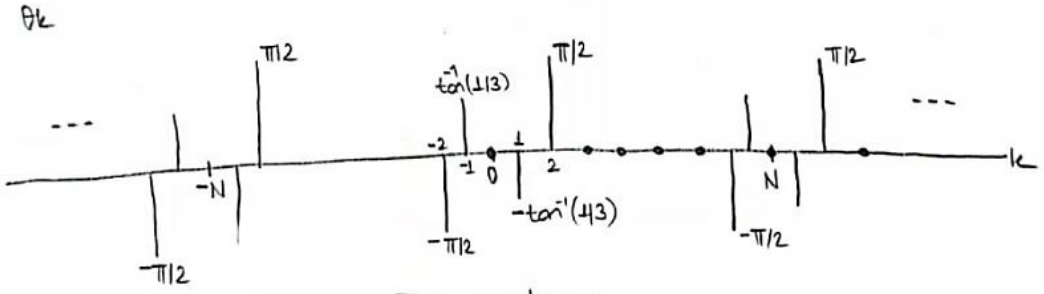
$\longrightarrow |x_2| = |x_{-2}| = 1/2$

$\theta_2 = \pi/2 \quad \theta_{-2} = -\pi/2$

$\Rightarrow x[n]$ işareti reel olduğundan $x_k^* = x_{-k}$'dır.

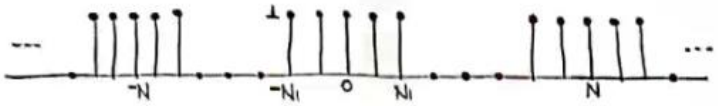


Genlik spektrumu



Faz spektrumu

Ex:



Ayrık zamanlı periyodik dikdörtgen dalga işaretinin FS katsayılarını bulunuz.

$$x_k = \frac{1}{N}\sum_{n=<N>} x[n]\,e^{-jk\omega_0 n} = \frac{1}{N}\sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n} = \frac{1}{N}\sum_{n=0}^{2N_1} e^{-jk\omega_0 (n-N_1)}$$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1}\sum_{n=0}^{2N_1} e^{-jk\omega_0 n} \quad \longleftarrow \text{Reminder: } \sum_{i=0}^{M} a^i = \frac{1-a^{N+1}}{1-a}$$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1}\left[\frac{1-e^{-jk\omega_0(2N_1+1)}}{1-e^{-jk\omega_0}}\right] \quad \longleftarrow \text{Reminder: } 1-e^{-j\theta} = e^{-j\theta/2}\left[e^{j\theta/2}-e^{-j\theta/2}\right]$$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1}\,\frac{e^{-jk\omega_0(N_1+1/2)}\left[e^{jk\omega_0(N_1+1/2)} - e^{-jk\omega_0(N_1+1/2)}\right]}{e^{-jk\omega_0/2}\left[e^{jk\omega_0/2}-e^{-jk\omega_0/2}\right]}$$

$$x_k = \frac{1}{N}\,\frac{\sin(k\omega_0(N_1+1/2))}{\sin(k\omega_0/2)}, \quad k \neq 0, \mp N, \mp 2N, \dots$$

$$x_k = \frac{2N_1+1}{N}, \quad k = 0, \mp N, \mp 2N, \dots$$

Parseval Teoremi

$$P_x = \frac{1}{N}\sum_{n=<N>} |x[n]|^2 = \sum_{k=<N>} |x_k|^2$$

Ayrık zamanlı periyodik bir işaretin ortalama gücü, bütün harmonik bileşenlerin (ki bunlar N tanedir) ortalama güçlerinin toplamına eşittir.

Ayrık Zamanlı Fourier Serisinin Özellikleri

$x[n] = x[n+N]$, $\omega_0 = \frac{2\pi}{N}$

$x[n] \xleftrightarrow{FS} x_k$

1) Çarpma: $x_1[n]\cdot x_2[n] \xleftrightarrow{FS} \sum_{l=<N>} x_{1,l}\cdot x_{2,k-l}$

İspat: $x_1[n]\,x_2[n] = \sum_{k=<N>} x_{1,k}\,e^{jk\omega_0 n}\sum_{l=<N>} x_{2,l}\,e^{jl\omega_0 n}$

$$= \sum_{k=0}^{N-1}\sum_{l=0}^{N-1} x_{1,k}\,x_{2,l}\,e^{j(k+l)\omega_0 n} = \sum_{k=0}^{N-1}\sum_{r=k}^{k+N-1} x_{1,k}\,x_{2,r-k}\,e^{jr\omega_0 n}$$

($k+l=r$)

$$= \sum_{r=<N>}\underbrace{\sum_{k=<N>} x_{1,k}\,x_{2,r-k}}_{d_r}\cdot e^{jr\omega_0 n}$$

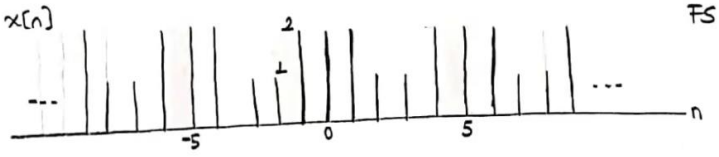
$$= \sum_{r=<N>} d_k\,e^{jr\omega_0 n}$$

2) İlk fark (first difference): $x[n]-x[n-1] \xleftrightarrow{FS} (1-e^{-jk\omega_0})x_k$

İspat: $x[n-1] = \sum_{k=<N>} x_k e^{jk\omega_0(n-1)} = e^{-jk\omega_0} \sum_{k=<N>} x_k e^{jk\omega_0 n} = e^{-jk\omega_0} x[n]$

$x[n]-x[n-1] = x[n]-e^{-jk\omega_0}x[n] = (1-e^{-jk\omega_0})x[n] \xleftrightarrow{FS} (1-e^{-jk\omega_0})x_k$

Ex:

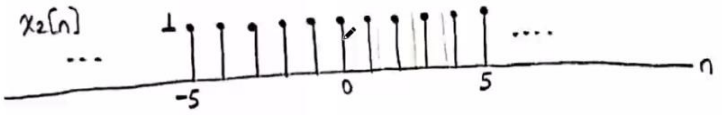


FS katsayılarını bulmuz.

N=5



⇒ $N_1=1$ ve $N=5$ olan periyodik dikdörtgen dalga



$x[n] = x_1[n]+x_2[n]$

$$x_{1,k} = \begin{cases} \frac{1}{N}\frac{\sin(k\omega_0(N_1+1/2))}{\sin(k\omega_0/2)}, & k \neq 0, \mp N, \mp 2N, \dots \\ \frac{2N_1+1}{N}, & k = 0, \mp N, \mp 2N \end{cases} \overset{\substack{N=5 \\ N_1=1}}{=} \begin{cases} \frac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}, & k \neq 0, \mp 5, \dots \\ 3/5, & k = 0, \mp 5, \mp 10, \dots \end{cases}$$

$$x_{2,k} = \frac{1}{N}\sum_{k=<N>} x_2[n]e^{jk\omega_0 n} \xrightarrow[N=5]{k=0} x_{2,0} = \frac{1}{5}\sum_{k=0}^{4} x_2[n] = 1 \Rightarrow x_{2,k} = \begin{cases} 1, & k = 0, \mp 5, \mp 10, \dots \\ 0, & k \neq 0, \mp 5, \mp 10, \dots \end{cases}$$

$x[n] = x_1[n]+x_2[n] \xleftrightarrow{FS} x_{1,k}+x_{2,k}$

$$x_k = \begin{cases} \frac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}, & k \neq 0, \mp 5, \mp 10, \dots \\ 8/5, & k = 0, \mp 5, \mp 10, \dots \end{cases}$$

Ex: Bir $x[n]$ işareti, aşağıda verilen özellikler ile belirlenebilir mi?

1) N=6 periyodu ile periyodiktir. Wo=2pi/N=pi/3

2) $\sum_{n=0}^{5} x[n] = 2$

3) $\sum_{n=2}^{7} (-1)^n x[n] = 1$

4) $x[n]$, yukarıdaki üç özelliği sağlayan işaretler arasında minimum güce sahiptir.

2) $x_k = \frac{1}{N}\sum_{n=<N>} x[n]e^{-jk\omega_0 n}$ olup $x_0 = \frac{1}{N}\sum_{n=<N>} x[n] \rightarrow Nx_0 = \sum_{n=<N>} x[n]$

$Nx_0 = 2 \rightarrow x_0 = 2/N = 1/3$

3) $Nx_k = \sum_{n=<N>} x[n]e^{-jk\omega_0 n} = 1$

$(-1)^n = e^{-j\pi n} \longrightarrow e^{-j\pi n} = e^{-jk\omega_0 n} \rightarrow k = \frac{\pi}{\omega_0} = \frac{\pi}{\frac{2\pi}{6}} = 3$

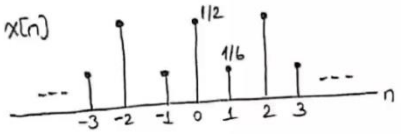
$x_3 = 1/6$

4) $P_x = \sum_{k=<N>} |x_k|^2 = \sum_{k=0}^{5} |x_k|^2$ ← $x_0 = 1/3$ ve $x_3 = 1/6$

Minimum güçlü olabilmesi için $x_1 = x_2 = x_4 = x_5 = 0$ seçilir.

$x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n} = x_0 + x_3 e^{j3\omega_0 n} = x_0 + x_3 e^{j\pi n} = \frac{1}{3} + \frac{1}{6}(-1)^n$

($\omega_0 = \pi/3$)



Ayrık zamanlı LTI sistemlerin periyodik işaretlere cevabı

$Az^n \longrightarrow \boxed{h[n]} \longrightarrow Az^n H(z)$, $H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k}$

$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$

$= \sum_{k=-\infty}^{\infty} h[k]Az^{n-k} = Az^n \underbrace{\sum_{k=-\infty}^{\infty} h[k]z^{-k}}_{\triangleq H(z)} = Az^n H(z) \rightarrow Az^n$: öz-fonksiyon

$\hookrightarrow H(z)$: özdeğer

$\Rightarrow z = e^{j\omega}$

$Ae^{j\omega n} \longrightarrow \boxed{h[n]} \longrightarrow Ae^{j\omega n} H(e^{j\omega})$

$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$: Frekans cevabı (frequency response)

$x[n] = \sum_{k=<N>} x_k e^{jk\omega_0 n} \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{k=<N>} x_k H(e^{jk\omega_0}) e^{jk\omega_0 n} = \sum_{k=<N>} y_k e^{jk\omega_0 n}$

$y_k \triangleq x_k H(e^{jk\omega_0})$

Ex: $h[n] = \alpha^n u[n]$, $-1 < \alpha < 1$ ve $x[n] = \cos\left(\frac{2\pi}{N} n\right)$ ise $y[n] = ?$

$$x[n] = \cos\left(\frac{2\pi}{N} n\right) = \cos(\omega_0 n) = \frac{1}{2}\left[e^{j\omega_0 n} + e^{-j\omega_0 n}\right] \longrightarrow x_1 = x_{-1} = \frac{1}{2}$$

$$\frac{2\pi}{N} = \omega_0$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} = \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n$$

Reminder: $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$, $|a| < 1$

$$= \frac{1}{1 - \alpha e^{-j\omega}}$$

$$y_k = x_k H(e^{jk\omega_0}) \longrightarrow y_1 = x_1 H(e^{j\omega_0}) = \frac{1}{2(1 - \alpha e^{-j\omega_0})}$$

$$\longrightarrow y_{-1} = x_{-1} H(e^{-j\omega_0}) = \frac{1}{2(1 - \alpha e^{j\omega_0})}$$

$$y[n] = \sum_{k=\langle N \rangle} y_k e^{jk\omega_0 n} = y_1 e^{j\omega_0 n} + y_{-1} e^{-j\omega_0 n}$$

$$y[n] = \frac{1}{2}\left(\frac{1}{1 - \alpha e^{-j\omega_0}}\right) e^{j\omega_0 n} + \frac{1}{2}\left(\frac{1}{1 - \alpha e^{j\omega_0}}\right) e^{-j\omega_0 n}$$

$$\frac{1}{1 - \alpha e^{-j\omega_0}} = r e^{j\theta}$$

$$y[n] = \frac{1}{2} r e^{j\theta} e^{j\omega_0 n} + \frac{1}{2} r e^{-j\theta} e^{-j\omega_0 n} = \frac{r}{2}\left[e^{j(\omega_0 n + \theta)} + e^{-j(\omega_0 n + \theta)}\right] = r\cos(\omega_0 n + \theta)$$

Ayrık zamanlı periyodik dikdörtgen dalga işaretinin FS katsayılarını bulunuz.

$$x_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=-N_1}^{N_1} e^{-jk\omega_0 n} = \frac{1}{N} \sum_{n=0}^{2N_1} e^{-jk\omega_0 (n - N_1)}$$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1} \sum_{n=0}^{2N_1} e^{-jk\omega_0 n}$$

Reminder: $\sum_{i=0}^{M} a^i = \frac{1 - a^{N+1}}{1-a}$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1} \left[\frac{1 - e^{-jk\omega_0 (2N_1 + 1)}}{1 - e^{-jk\omega_0}}\right]$$

Reminder: $1 - e^{-j\theta} = e^{-j\theta/2}\left[e^{j\theta/2} - e^{-j\theta/2}\right]$

$$x_k = \frac{1}{N} e^{jk\omega_0 N_1} \frac{e^{-jk\omega_0 (N_1 + 1/2)}\left[e^{jk\omega_0 (N_1 + 1/2)} - e^{-jk\omega_0 (N_1 + 1/2)}\right]}{e^{-jk\omega_0/2}\left[e^{jk\omega_0/2} - e^{-jk\omega_0/2}\right]}$$